OS Practical No. 8

Aim :

A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Least recently used page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3).

Code:

```
#include <stdio.h>

int main() {
   int frames[3];
   int pages[] = {1,2,3,4,1,2,5,1,2,3,4,5};
   int time[3];
   int i, j, k;
   int hit = 0, fault = 0;
   int n = 12, frames_count = 3;
   int counter = 0, flag;

   // initialize frames
   for(i = 0; i < frames_count; i++)
      { frames[i] = -1;
   }

   for(i = 0; i < n; i++)
      { flag = 0;

      // check page hit
      for(j = 0; j < frames_count; j++)
         { if(frames[j] == pages[i]) {
            hit++; counter++;
            time[j] = counter;
            flag = 1;
            break;
         }
      }
```

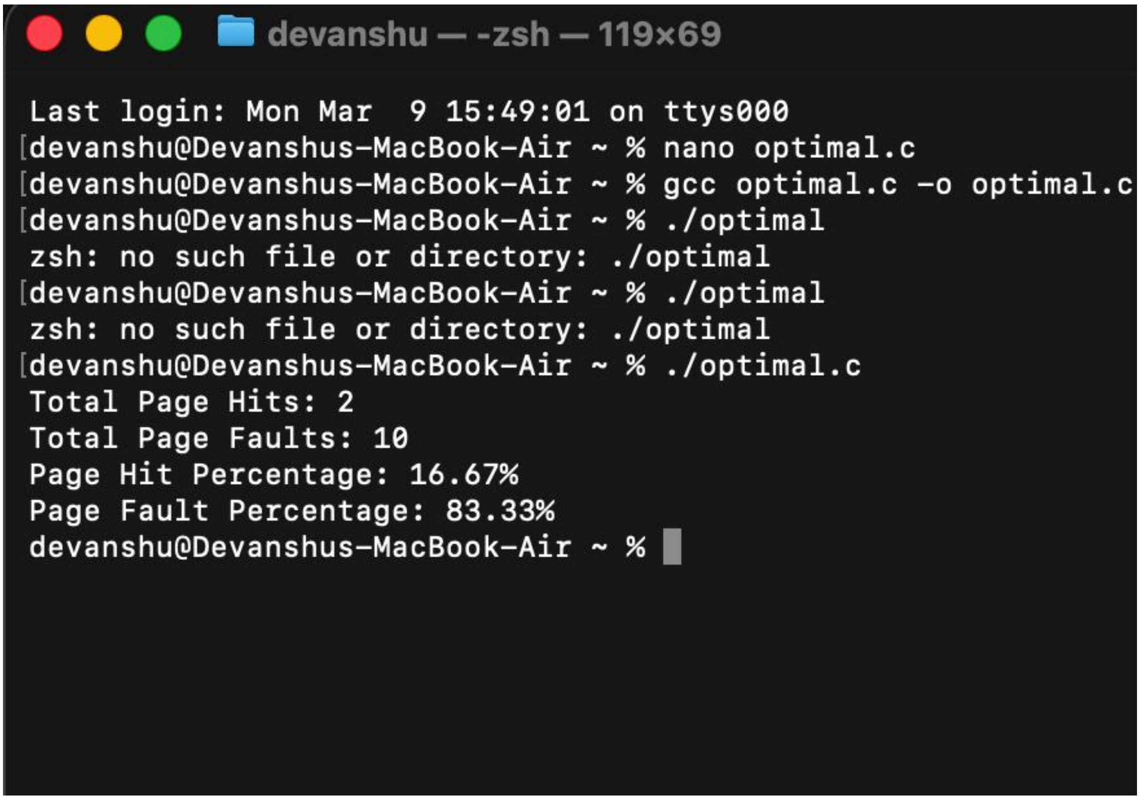
```
        // page fault
        if(flag == 0) {
            int min = 0;

            for(j = 1; j < frames_count; j++)
                { if(time[j] < time[min])
                    min = j;
            }
frames[min] = pages[i];
            counter++;
            time[min] = counter;
            fault++;
        }
    }
float hit_percent = (hit * 100.0) / n;
    float fault_percent = (fault * 100.0) / n;

    printf("Total Page Hits: %d\n", hit);
    printf("Total Page Faults: %d\n", fault);
    printf("Page Hit Percentage: %.2f%%\n", hit_percent);
    printf("Page Fault Percentage: %.2f%%\n", fault_percent);

    return 0;
}
```

Output :



```c
#include <stdio.h>

int main() {
    int frames[3];
    int pages[] = {1,2,3,4,1,2,5,1,2,3,4,5};
    int time[3];
    int i, j, k;
    int hit = 0, fault = 0;
    int n = 12, frames_count = 3;
    int counter = 0, flag;

    // initialize frames
    for(i = 0; i < frames_count; i++) {
        frames[i] = -1;
    }

    for(i = 0; i < n; i++) {
        flag = 0;

        // check page hit
        for(j = 0; j < frames_count; j++) {
            if(frames[j] == pages[i]) {
                hit++;
                counter++;
                time[j] = counter;
                flag = 1;
                break;
            }
        }

        // page fault
        if(flag == 0) {
            int min = 0;

            for(j = 1; j < frames_count; j++) {
                if(time[j] < time[min])
                    min = j;
            }

            frames[min] = pages[i];
            counter++;
            time[min] = counter;
            fault++;
        }
    }

    float hit_percent = (hit * 100.0) / n;
    float fault_percent = (fault * 100.0) / n;
```

```
^G Get Help    ^O WriteOut    ^R Read File   ^Y Prev Pg     ^K Cut Text    ^C Cur Pos
^X Exit        ^J Justify     ^W Where is    ^V Next Pg     ^U UnCut Text  ^T To Spell
```